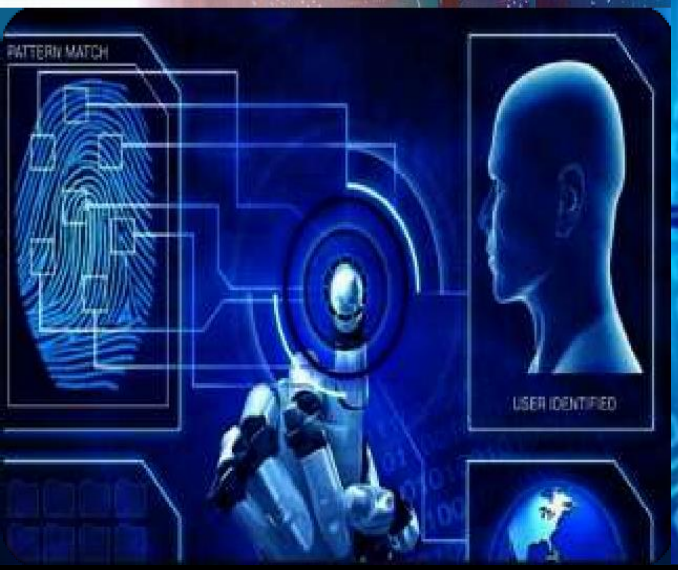




BIOMETRICS

BIOMETRICS



Dr. Gour Vishwavidyalaya, Sagar, M.P.

Identification Simplified



What we need ?



What we need ?

a method to **identify individuals** which is -



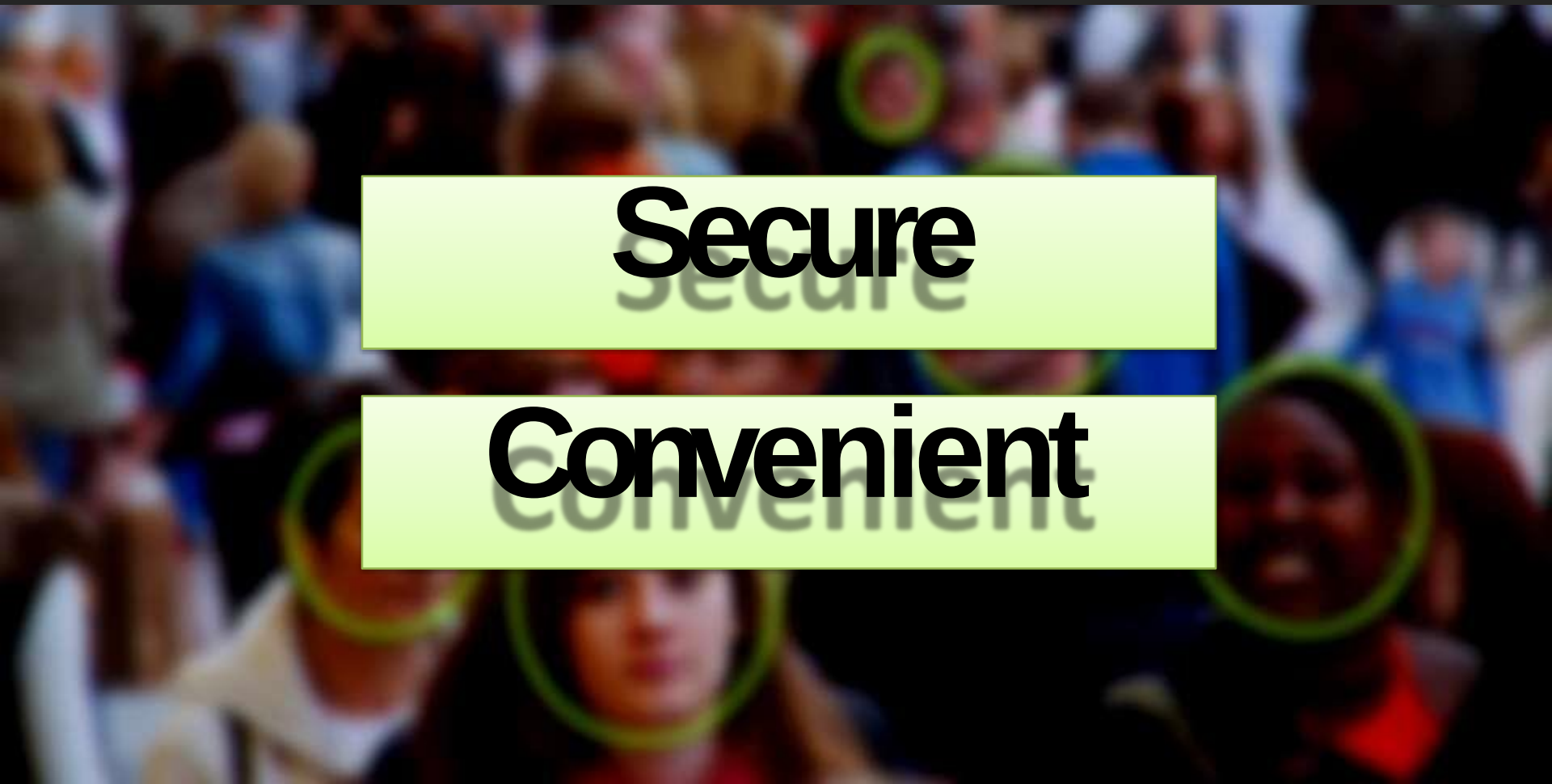
What we need ?

a method to **identify individuals** which is -



What we need ?

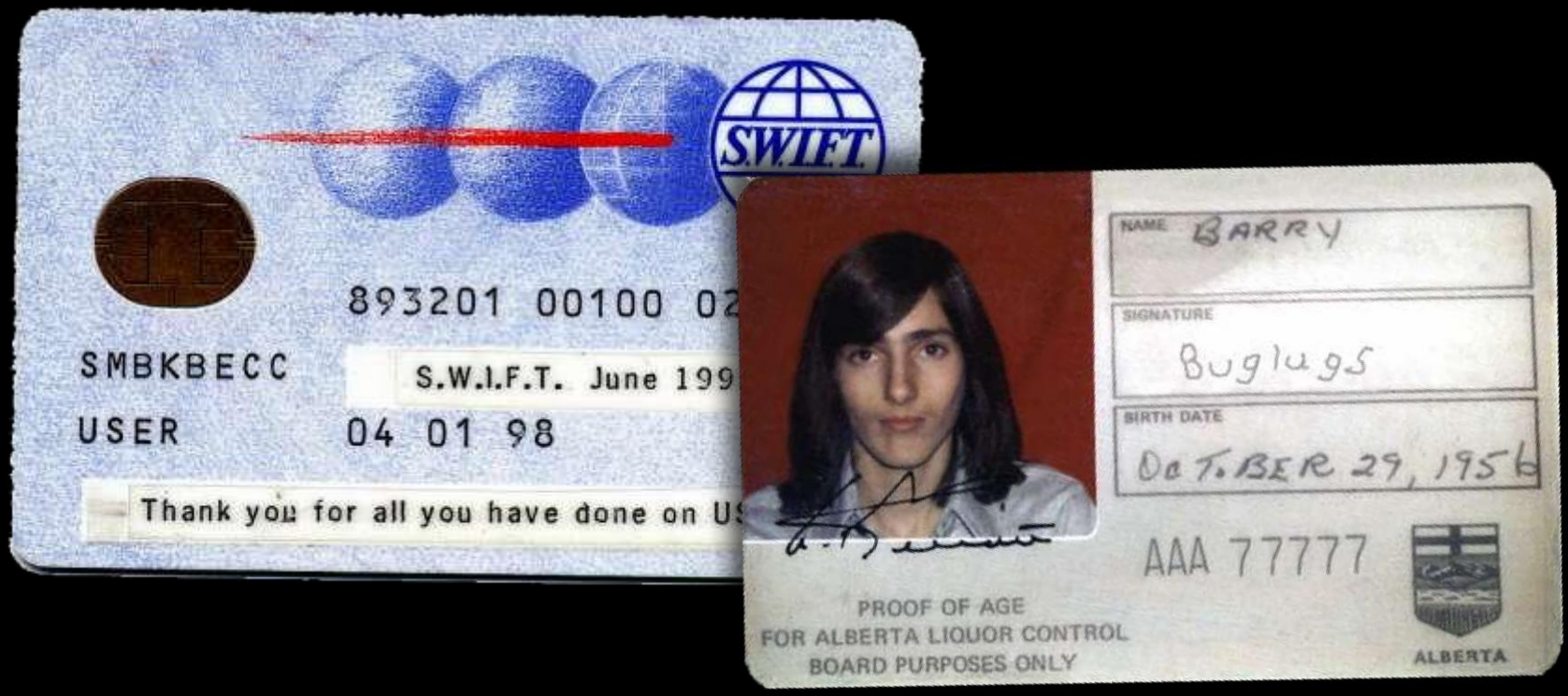
a method to **identify individuals** which is -



How do we do that ?

How do we do that ?

ID Cards & Smart Cards



How do we do that ?

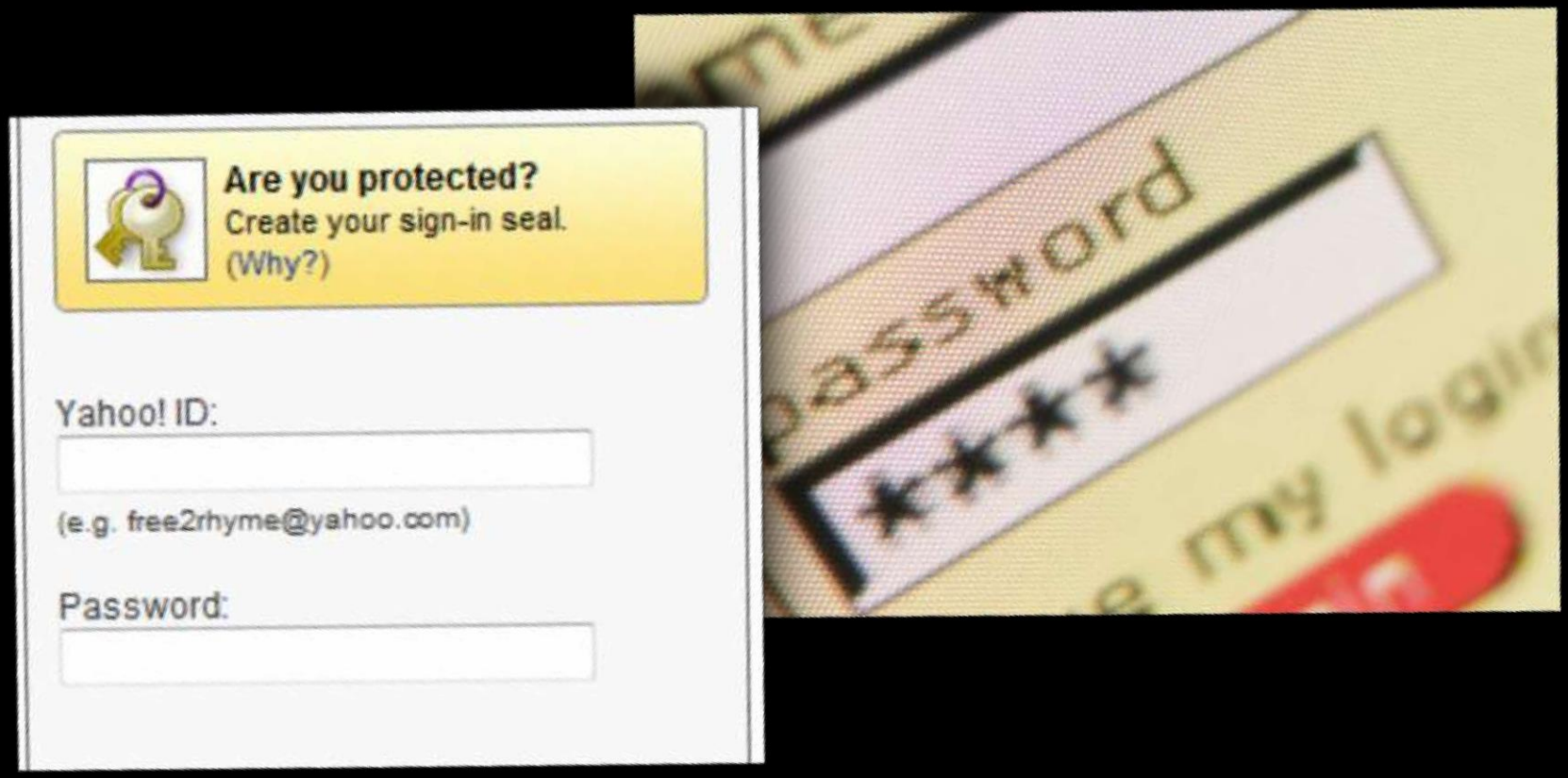
RFID Cards



- RFID – Radio Frequency Identification

How do we do that ?

Passwords



How do we do that ?

Wrist Bands



How do we do that ?

Wrist Bands

Passwords

ID Cards & Smart Cards

RFID Cards

How do we do that ?

Wrist Bands

Passwords

ID Cards & Smart Cards

RFID Cards



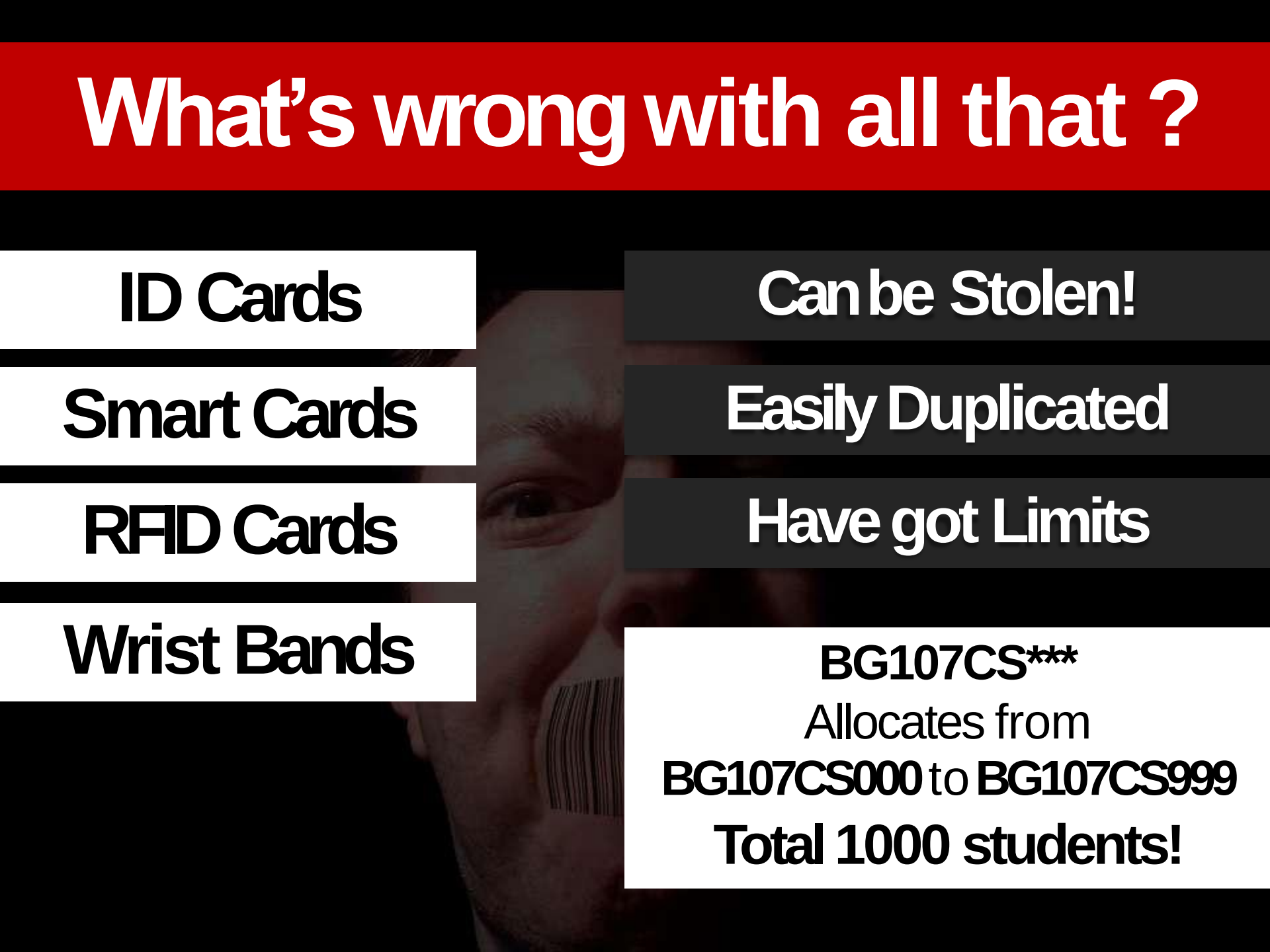
Wow! So many ways to identify people! **what could be wrong?**

What's wrong with all that ?



What's wrong with all that ?

- ID Cards
- Smart Cards
- RFID Cards
- Wrist Bands



- Can be Stolen!
- Easily Duplicated
- Have got Limits

BG107CS*

Allocates from

BG107CS000 to **BG107CS999**

Total 1000 students!

What's wrong with all that ?



What's wrong with all that ?

Passwords



What's wrong with all that ?

Passwords

Easily Forgotten



What's wrong with all that ?

Passwords

Easily Forgotten

Needs to be Changed



What's wrong with all that ?

Passwords

Easily Forgotten

Needs to be Changed



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MARCH 26, 2007

Up Front

Edited by Deborah Stead

STORY TOOLS

The Password Is "Expensive"

It typically costs a company $10 to $13 to reset an employee password, according to Forrester Research (FORR)—adding up, in some cases, to millions in annual expenditures as passwords get more complex. "It's always been a problem, but it's gotten a lot worse," says Jonathan Penn, the Cambridge (Mass.) firm's research director.

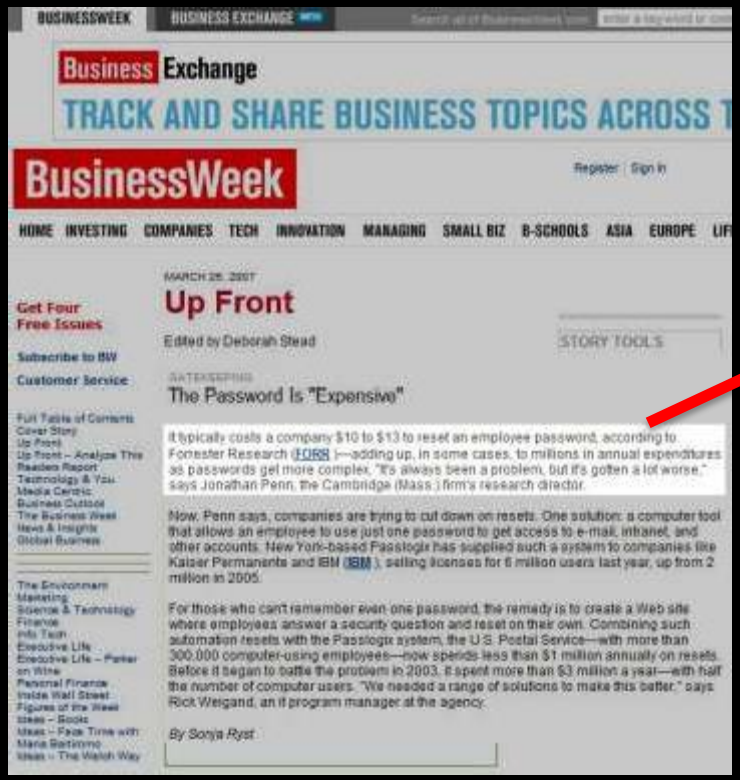
Now, Penn says, companies are trying to cut down on resets. One solution: a computer tool that allows an employee to use just one password to get access to e-mail, intranet, and other accounts. New York-based Passlogix has supplied such a system to companies like Kaiser Permanente and IBM (IBM), selling licenses for 6 million users last year, up from 2 million in 2005.

For those who can't remember even one password, the remedy is to create a Web site where employees answer a security question and reset on their own. Combining such automation resets with the Passlogix system, the U.S. Postal Service—with more than 300,000 computer-using employees—now spends less than $1 million annually on resets. Before it began to battle the problem in 2003, it spent more than $3 million a year—with half the number of computer users. "We needed a range of solutions to make this better," says Rick Weigand, an IT program manager at the agency.

By Sonja Ryst

What's wrong with all that ?

Passwords


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Business Exchange

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STORY TOOLS

GATEKEEPING

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By Sonya Ryst

Easily Forgotten

Needs to be Changed

The Password Is "Expensive"

It typically **costs** a company **$10 to $13 to reset an employee password**, according to **Forrester Research**

The answer to all this…



The answer to all this…



Lies Within Us!

In the form of Biometrics

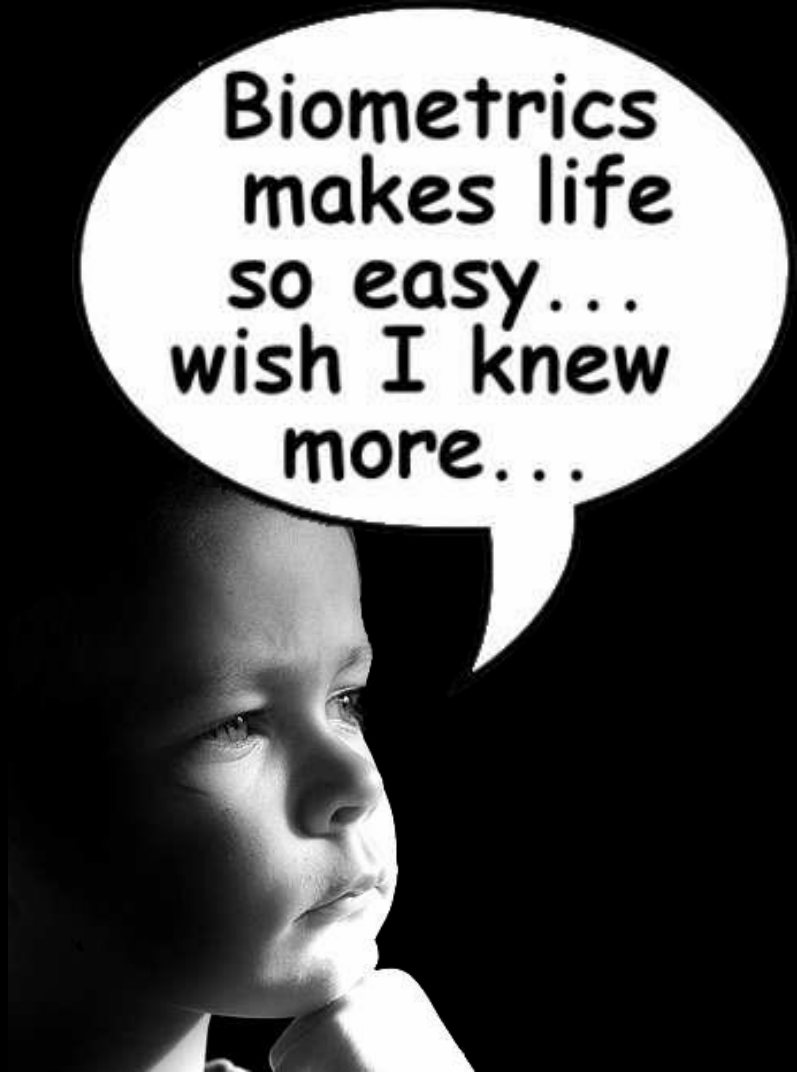
In the form of **Biometrics**



Biometrics is the study of **computerized methods** to **identify a person** by their **unique physical** or **behavioural** characteristics

In the form of Biometrics





INTRODUCTION

- Biometrics is the term derived form Greek word "***biometron***" means **Bio= life** & **Metron= to measure**.
- Biometrics defined as an tool for the **identification and verification** of the person on the basis of **physiological and/or behavioral** characteristics of the person.

BIOMETRIC TRAITS



Physiological Characteristics (Body Parts)

- Fingerprints
- Hand geometry
- Facial Recognition
- Iris Recognition
- Retina Recognition
- DNA
- Vein Pattern
- Skin Spectroscopy

- Voice Recognition
- Dynamic Signature Analysis
- Keystrokes Analysis
- Gait Pattern Analysis

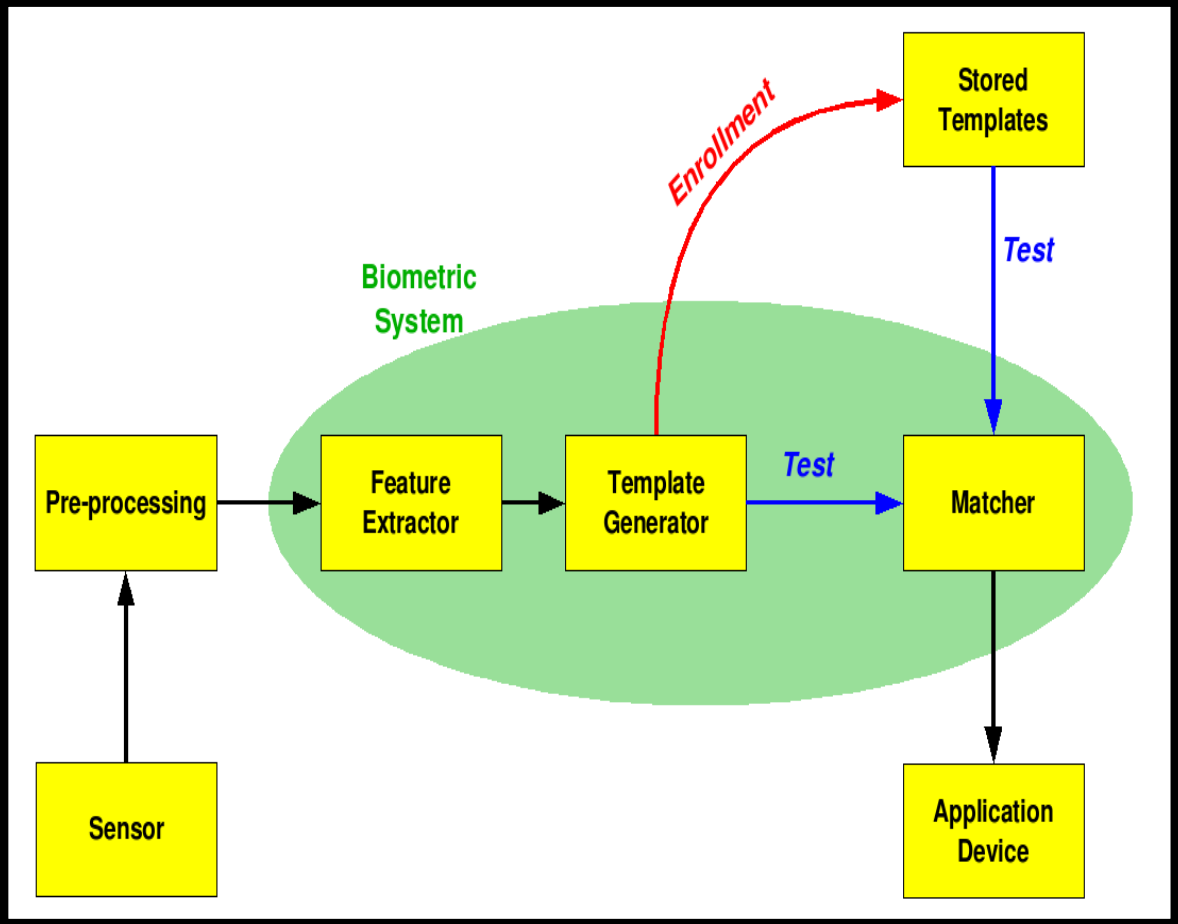


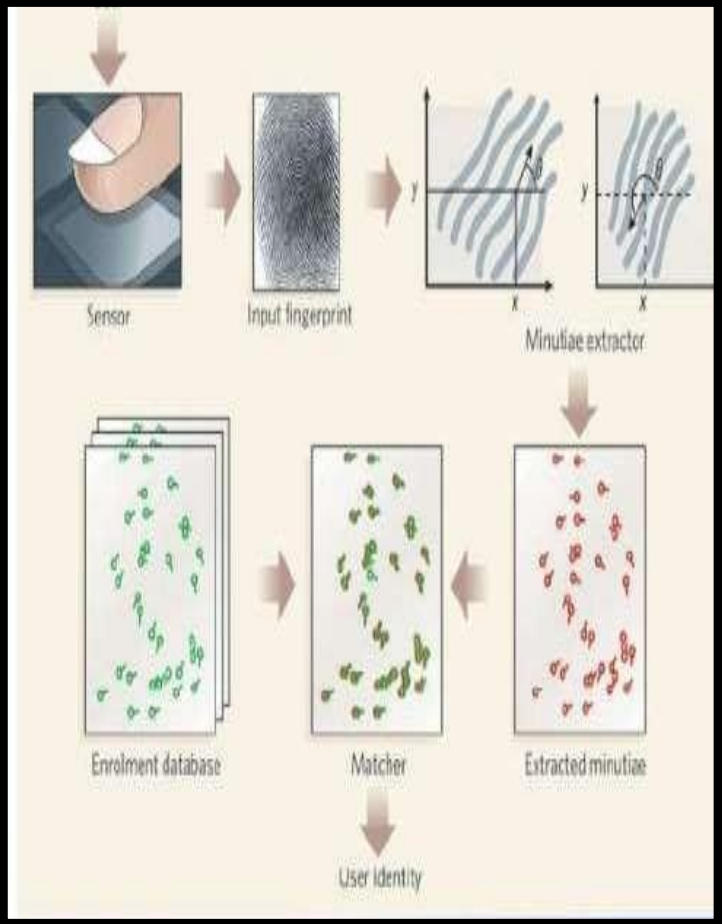


BASIC CHARACTERISTICS OF BIOMETRICS

- **Universality** : Describes how commonly biometrics found in each individual.
- **Uniqueness** : How well the biometrics separates from one individual to another.
- **Permanence** : Measure how well a biometrics resist again.
- **Collectability** : Explain how it is easy to acquire a biometrics for measurement.
- **Performance** : Indicate the accuracy, speed and robustness of the system capturing the biometrics.
- **Acceptability** : Indicate the degree of approval of a technology by the public in everyday life.
- **Circumvention** : It indicates how hard it is to fool the authentication system.

Working of Basic Biometric Model





What's in store for you?

What's in store for you?

Iris Recognition

Voice Recognition

Face Recognition

Fingerprint Authentication

Other Types

Iris Recognition

Iris Recognition

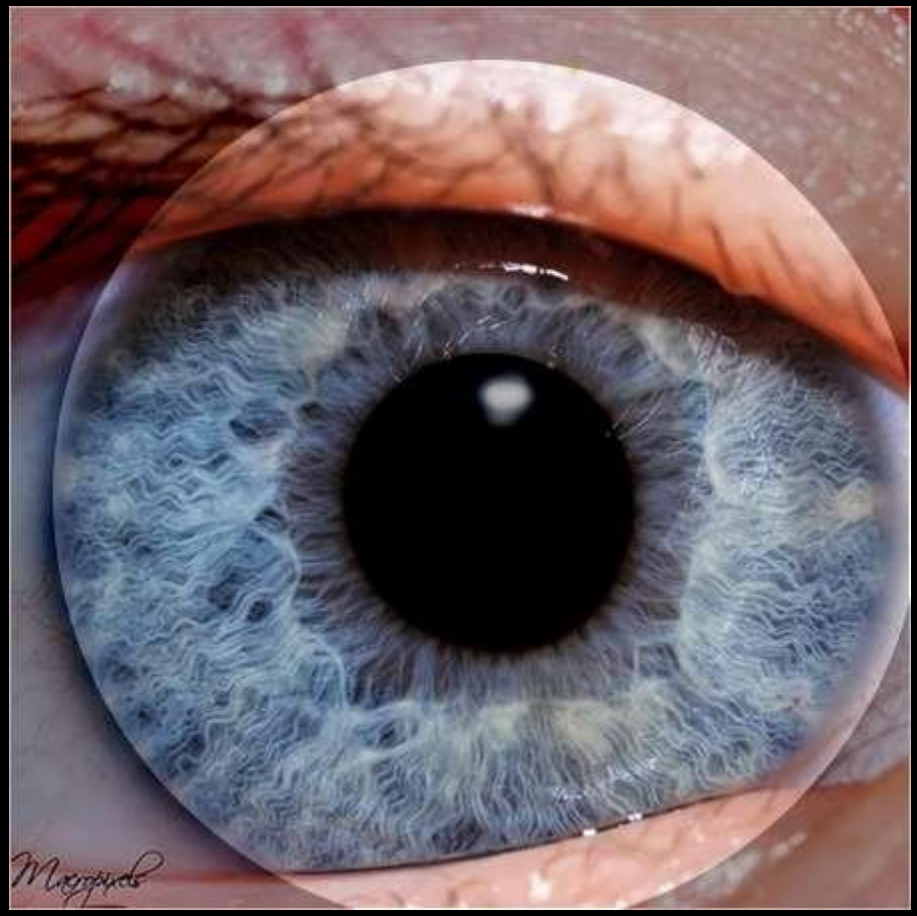
The Basics



Iris Recognition

The Basics

- Colored portion of the eye
- Unique
- Single enrollment for a lifetime
- Can be used with glasses or contacts



Iris Recognition

The Process

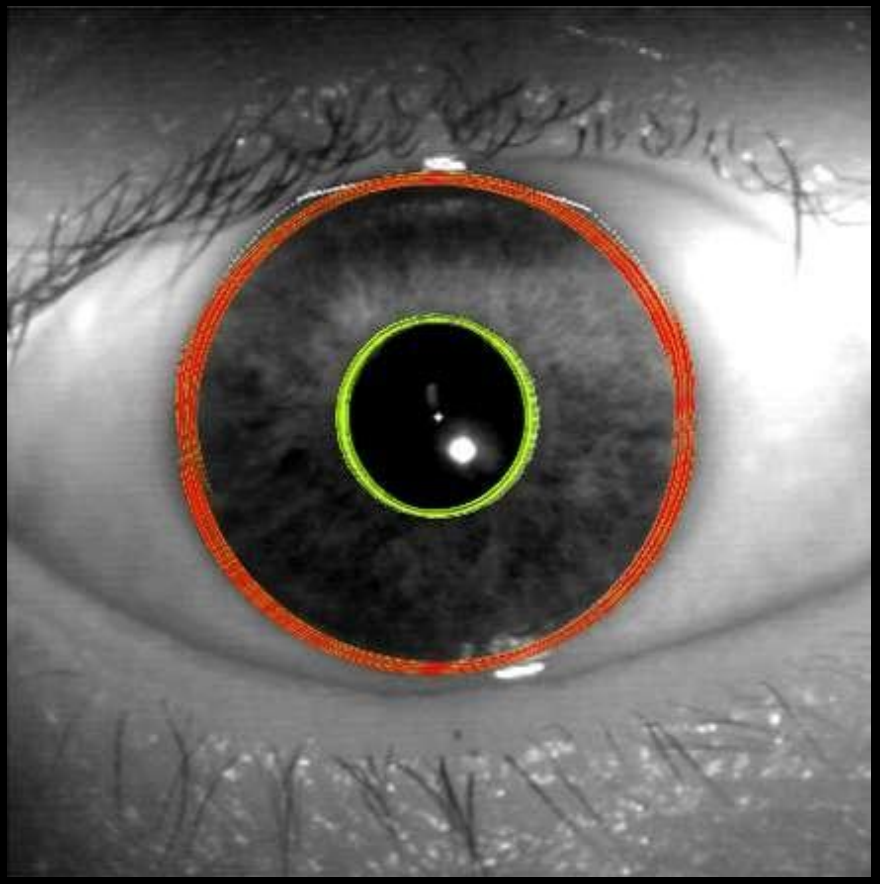
1. Image Acquisition



Iris Recognition

The Process

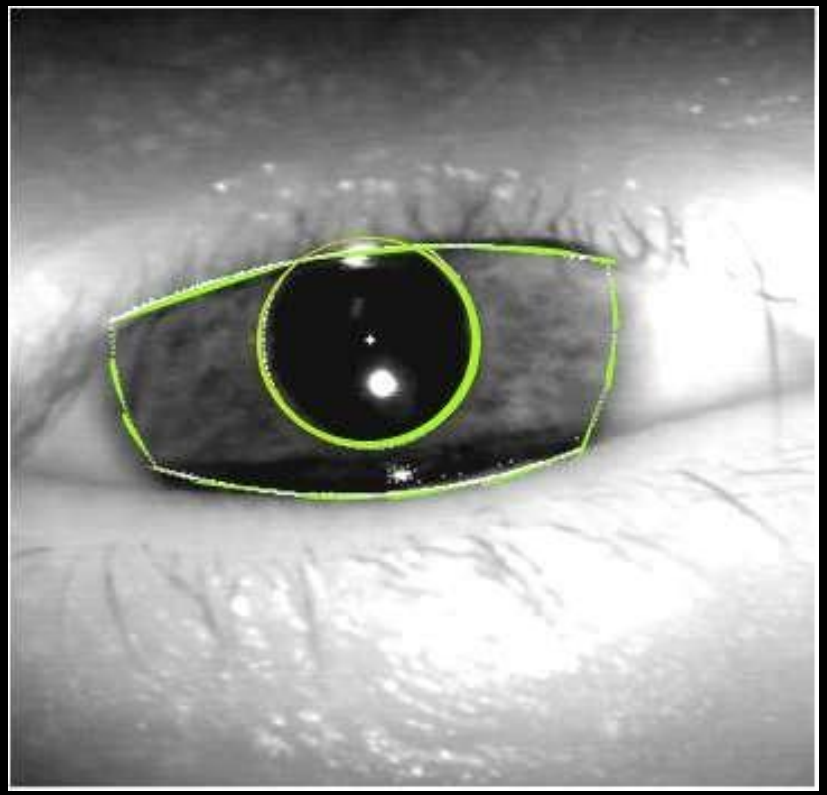
1. Image Acquisition
2. Iris Definition



Iris Recognition

The Process

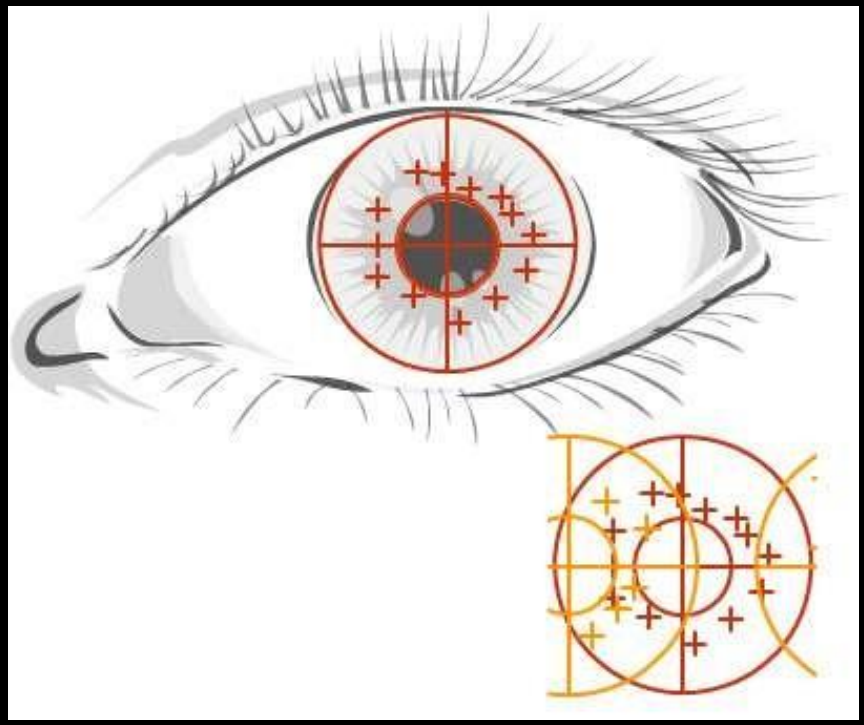
1. Image Acquisition
2. Iris Definition
3. Field Optimization



Iris Recognition

The Process

1. Image Acquisition
2. Iris Definition
3. Field Optimization
4. Image Analysis

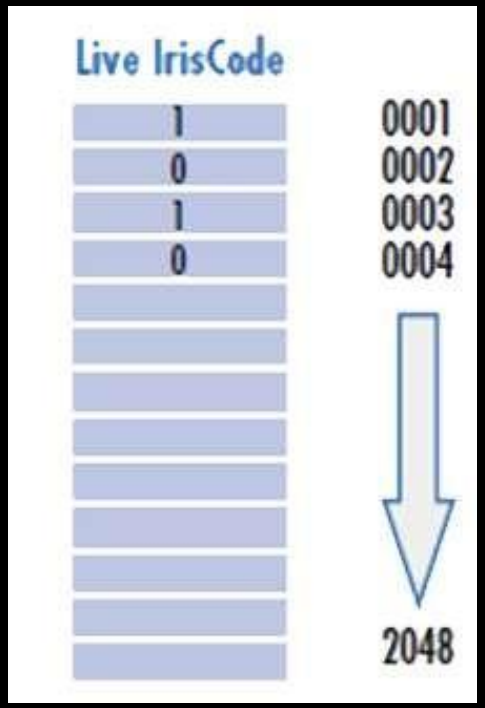


Iris Recognition

Iris Code Comparison

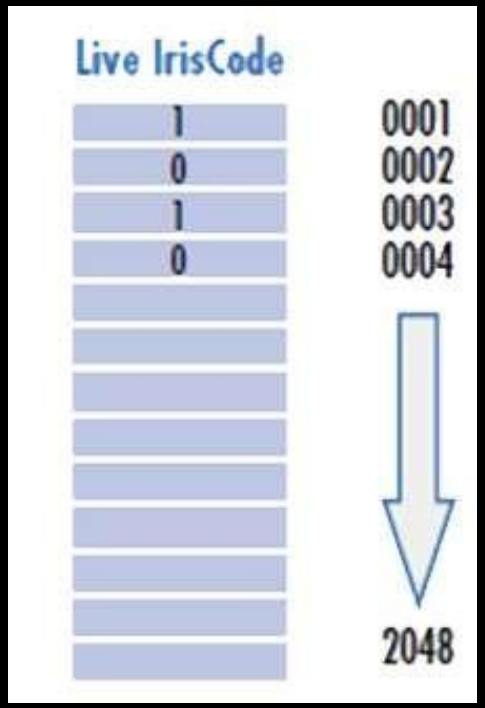
Iris Recognition

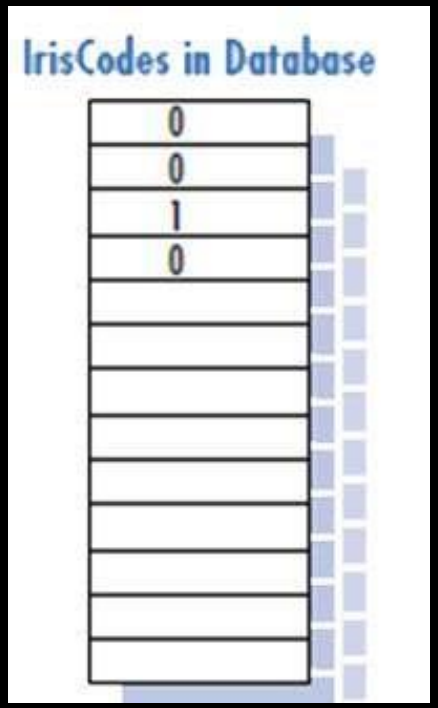
Iris Code Comparison



Iris Recognition

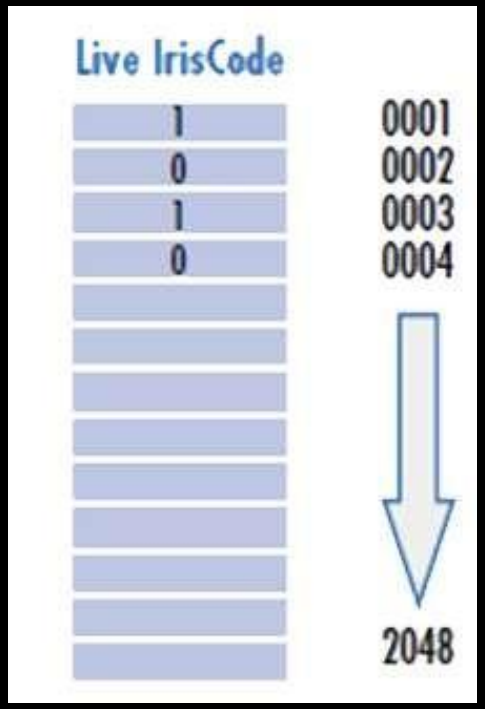
Iris Code Comparison

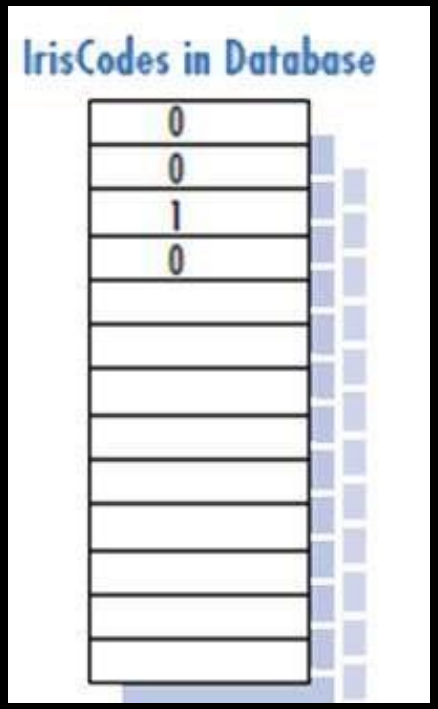


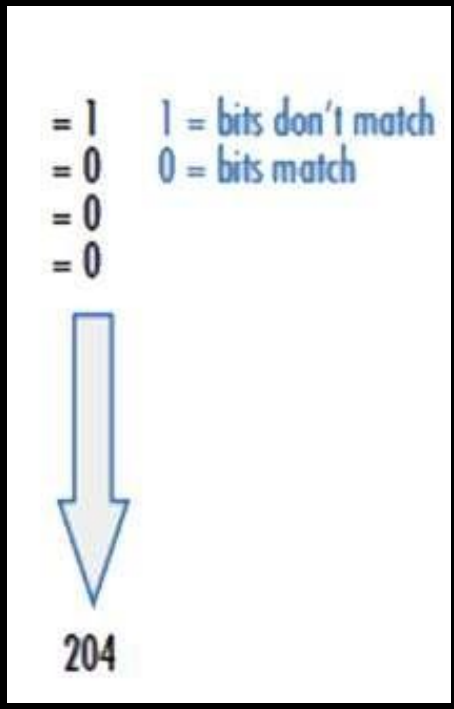


Iris Recognition

Iris Code Comparison







Mismatch = (204/2048)*100 = 10%

Iris Recognition

It's Fool-Proof

- 65.8% match
- Un-modifiable
- Pupil Dilation
- No replacement for a live eye



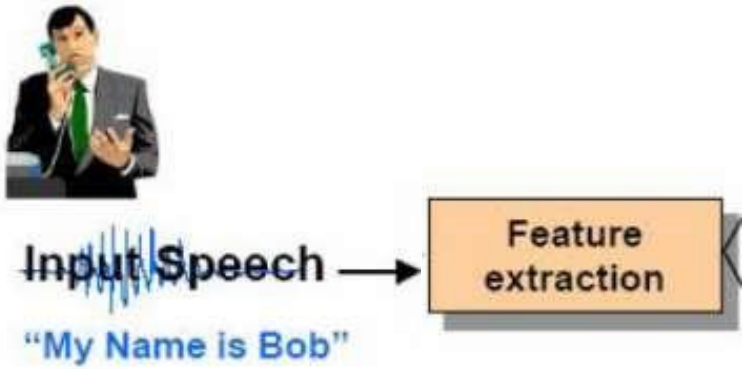
Voice Recognition



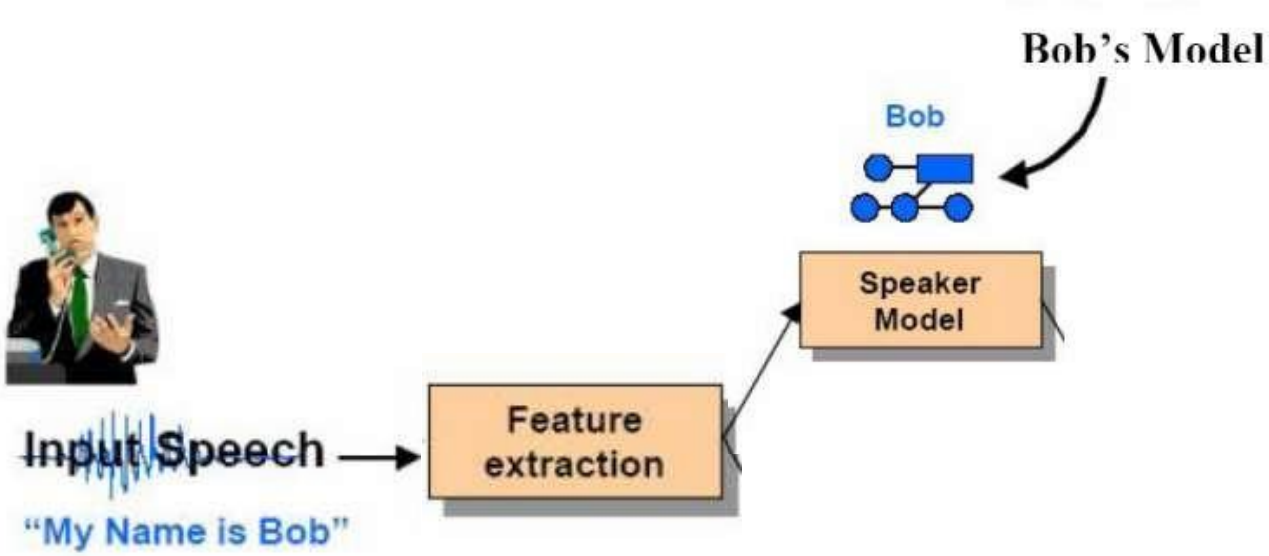
Input Speech →

"My Name is Bob"

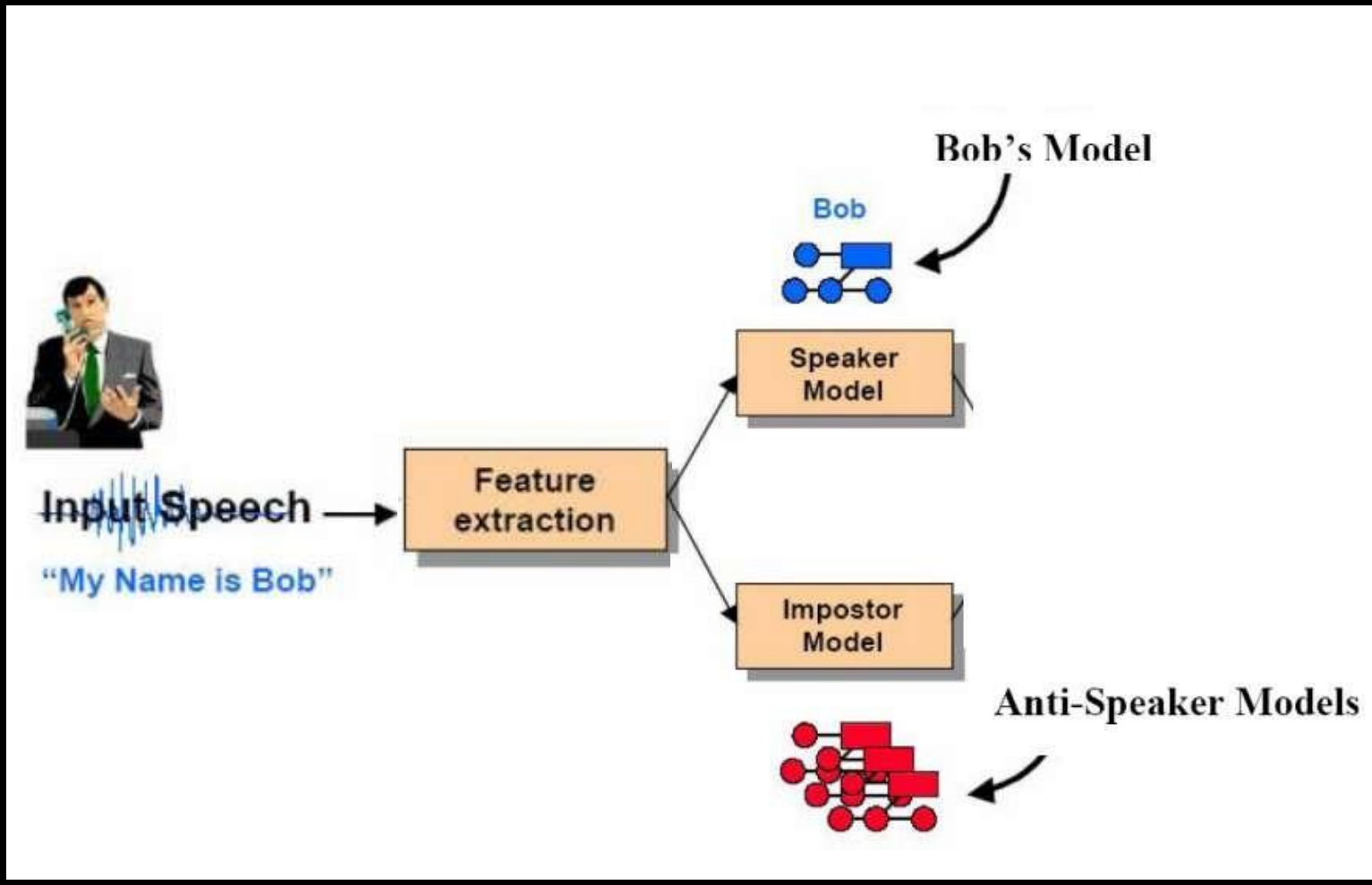
Voice Recognition



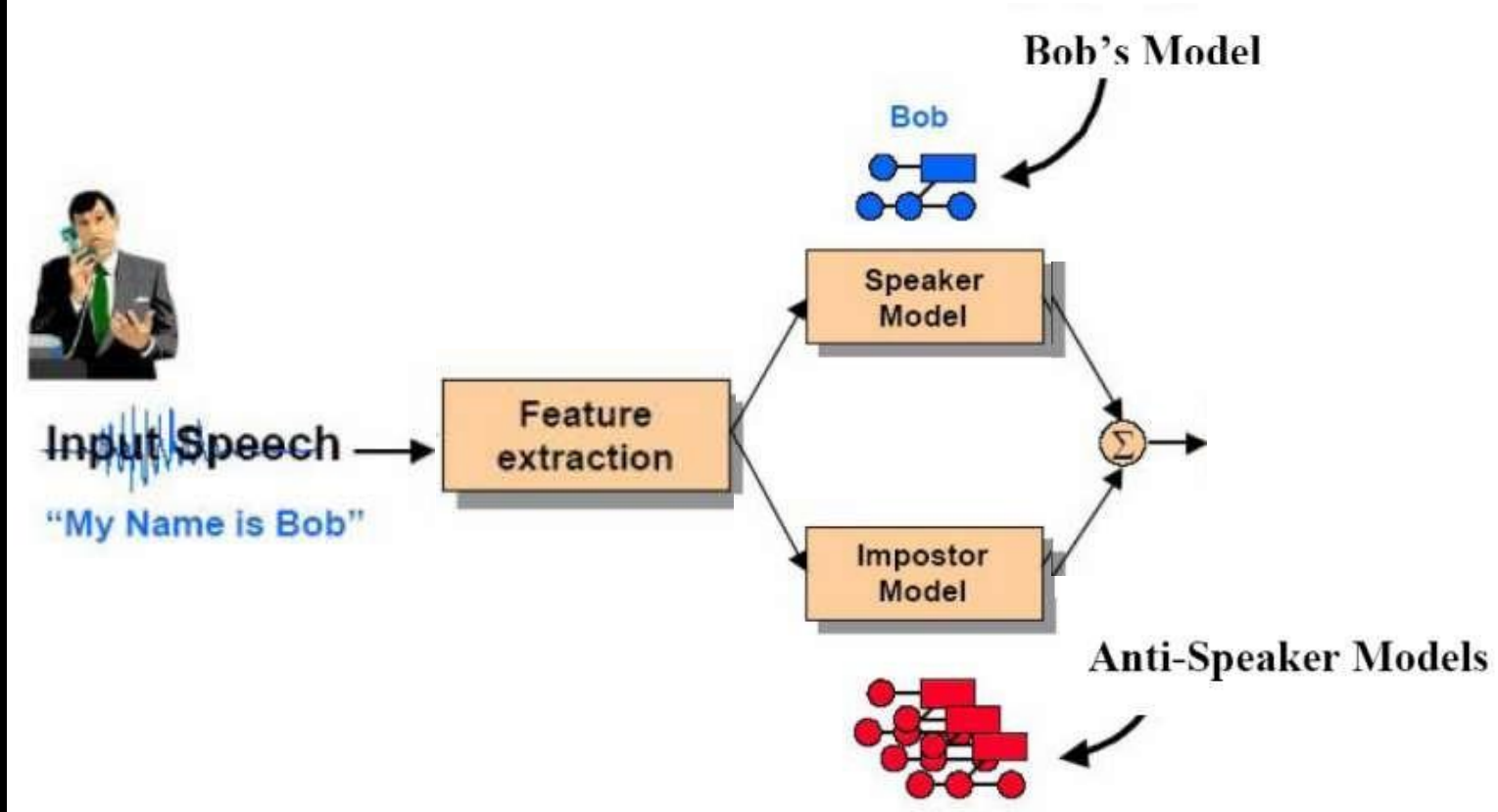
Voice Recognition



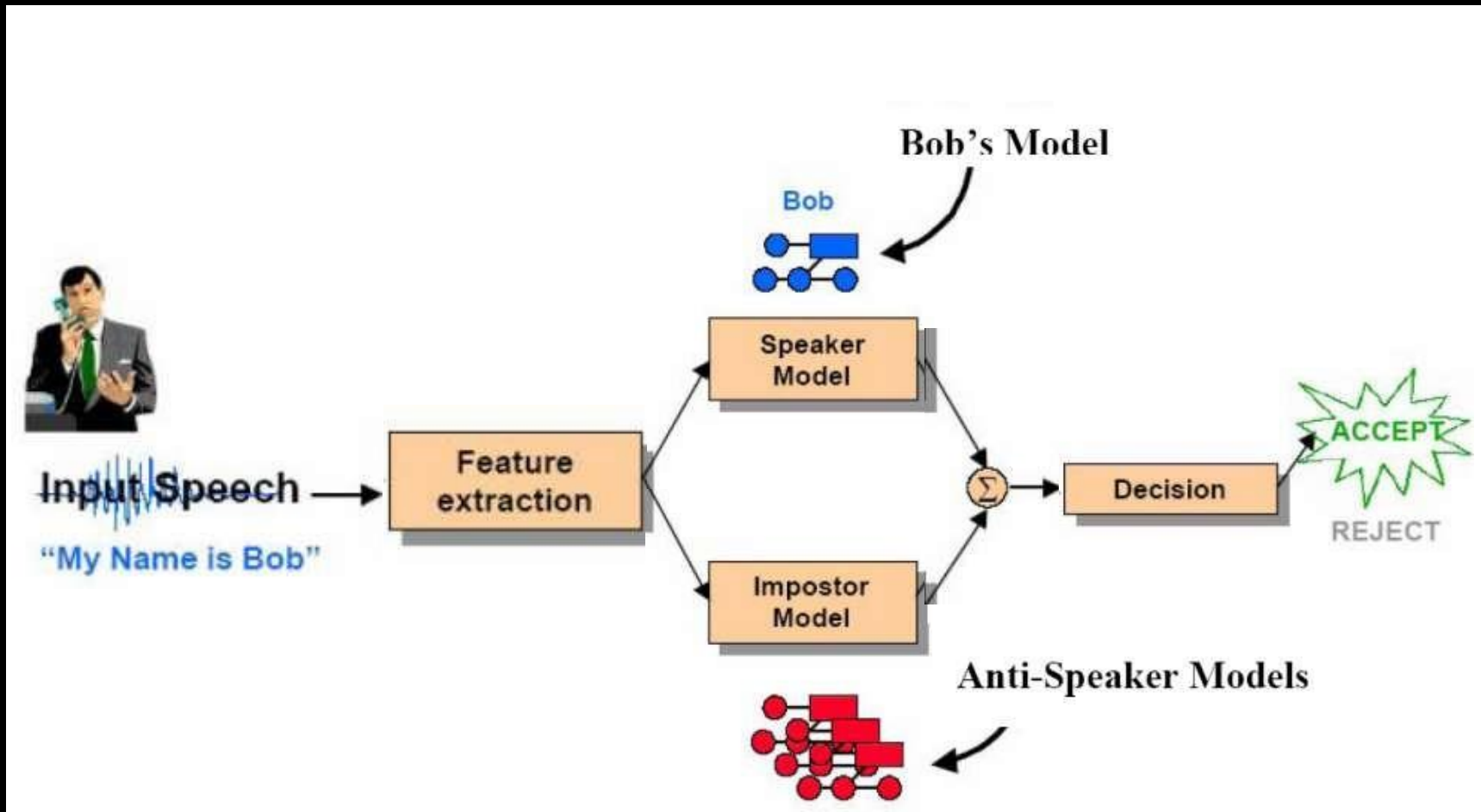
Voice Recognition



Voice Recognition



Voice Recognition



Voice Recognition

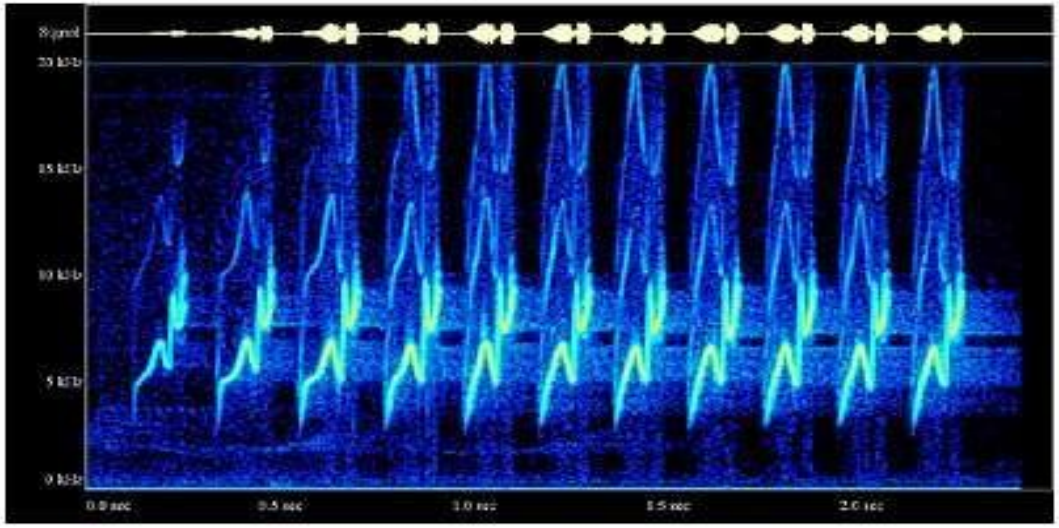
Approach

Voice Recognition

Approach

Waveform

Frequency Domain



Time

Voice Recognition

How is it done ?

Voice Recognition

How is it done ?

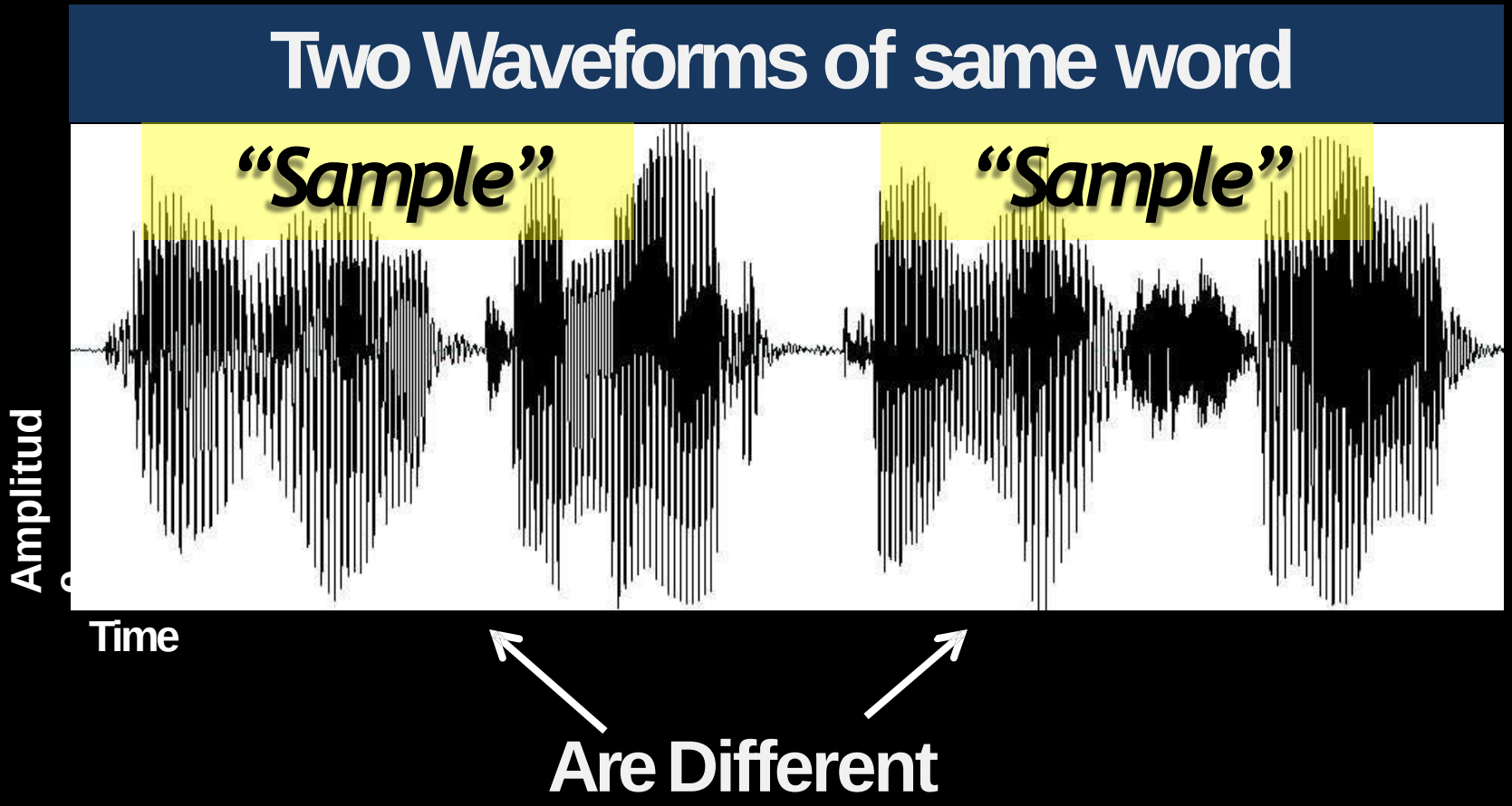
Two Waveforms of same word



- **Waveform** - the shape of a wave illustrated graphically
- **Amplitude** - displacement of a periodic wave

Voice Recognition

How is it done ?



Voice Recognition

How is it done ?

But the Frequency-Time plots



- **Frequency** - The number of occurrences of an event within a given interval

Voice Recognition

How is it done ?

But the Frequency-Time plots



Look Similar

Voice Recognition

Applications

Voice Recognition

Applications

Face Recognition

Face Recognition

Features

- Facial Features are Unique

Face Recognition

Features

- Facial Features are Unique
- Passive Biometrics

Face Recognition

Features

- Facial Features are Unique
- Passive Biometrics
- Already Widespread

Face Recognition

Features

- Facial Features are Unique
- Passive Biometrics
- Already Widespread
- Inexpensive

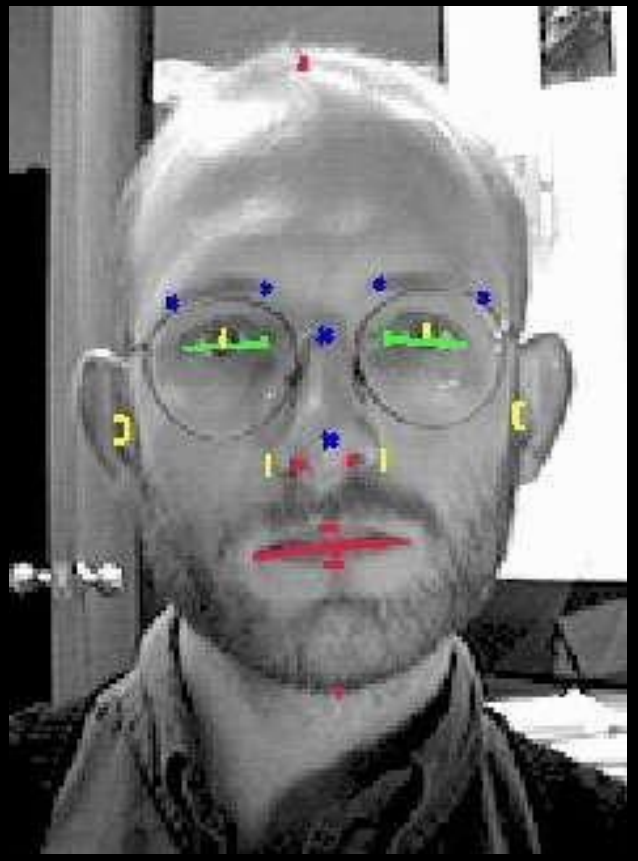
Face Recognition

Categories

Face Recognition

Categories

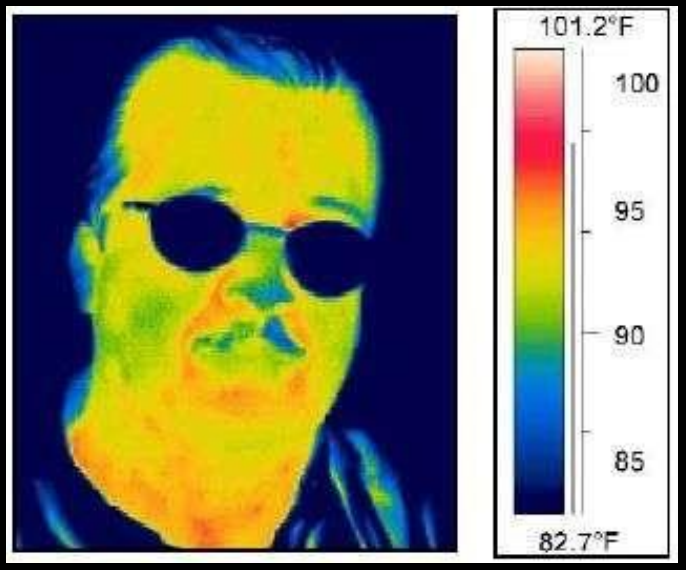
1. Face Geometry



Face Recognition

Categories

1. Face Geometry
2. Facial Thermograms



Face Recognition

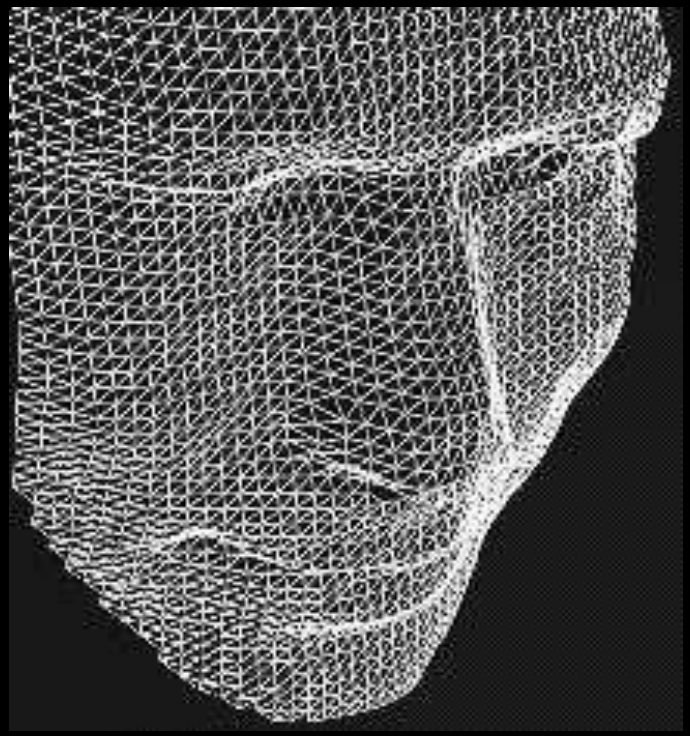
Categories

1. Face Geometry
2. Facial Thermograms
3. Eigen Face Method

Face Recognition

Categories

1. Face Geometry
2. Facial Thermograms
3. Eigen Face Method
4. Template Based

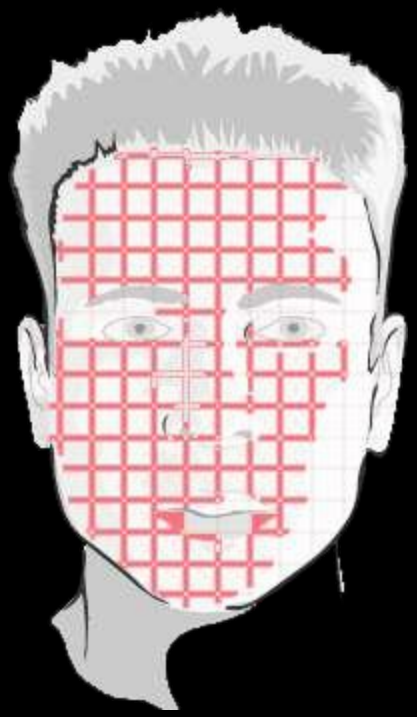


Face Recognition

Process

Face Recognition

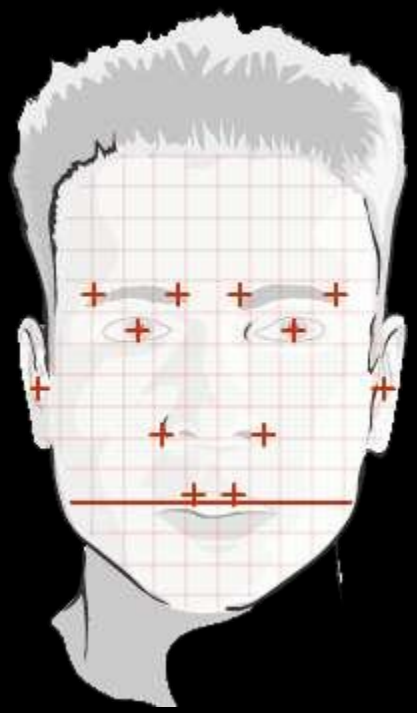
Process



Detection

Face Recognition

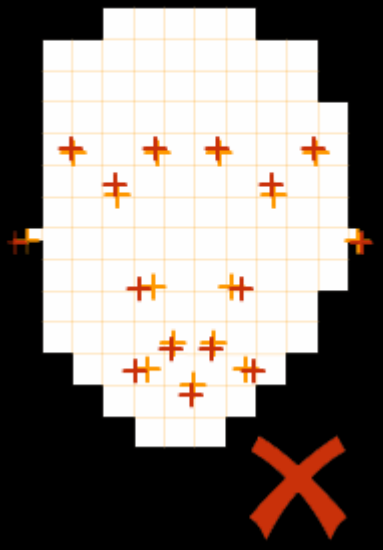
Process



Alignment & Measurement

Face Recognition

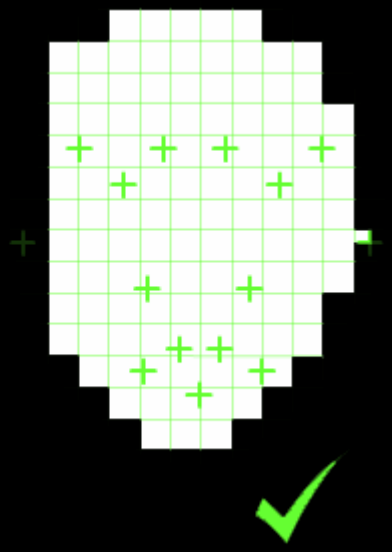
Process



Representation & Matching

Face Recognition

Process



Identity Confirmed

Face Recognition

Identification

WHO AM I?

Face Recognition

Verification

AM I WHO I SAY I AM ?

Fingerprint Authentication

Fingerprint Authentication

Features

Fingerprint Authentication

Features

- Ridges, Valleys

Fingerprint Authentication

Features

- Ridges, Valleys



Fingerprint Authentication

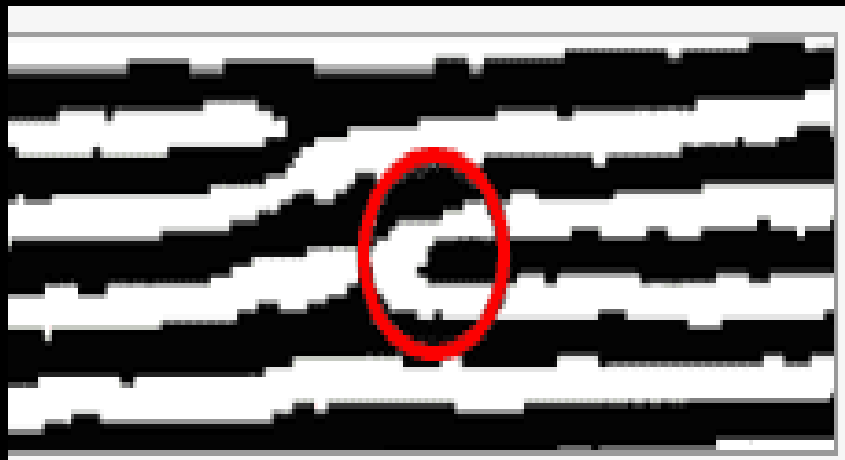
Features

- Ridges, Valleys
- Dark lines - Ridges
- Brighter lines - Valleys

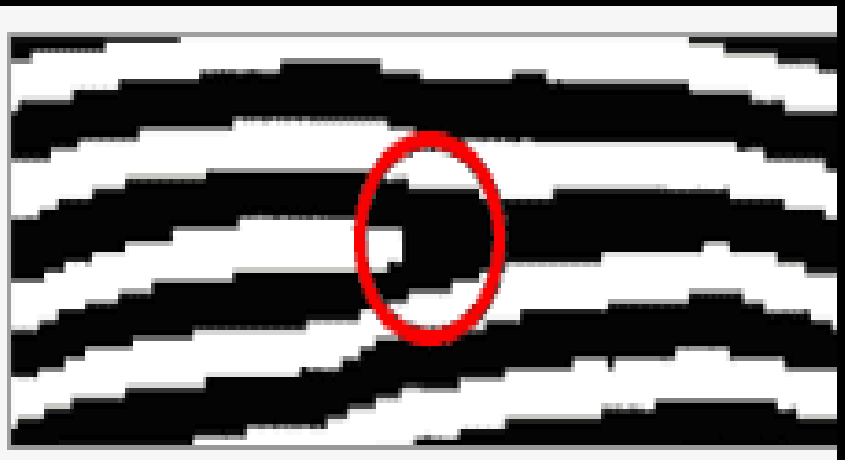


Fingerprint Authentication

Discriminating Information



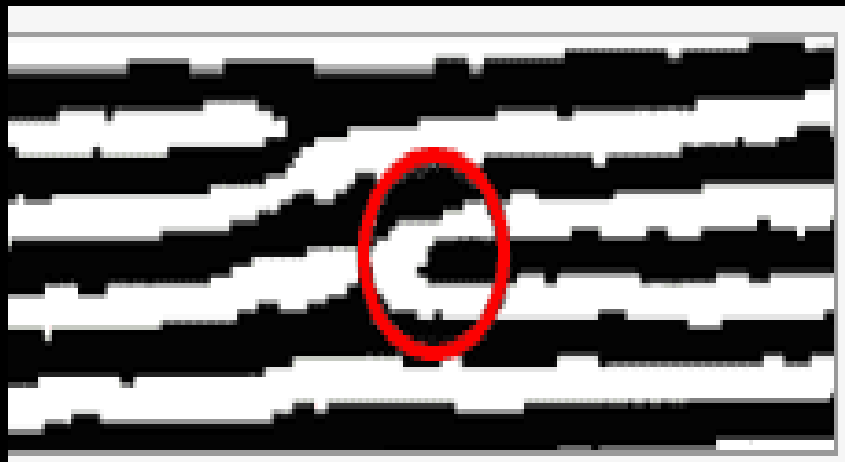
a. Ridge Ending



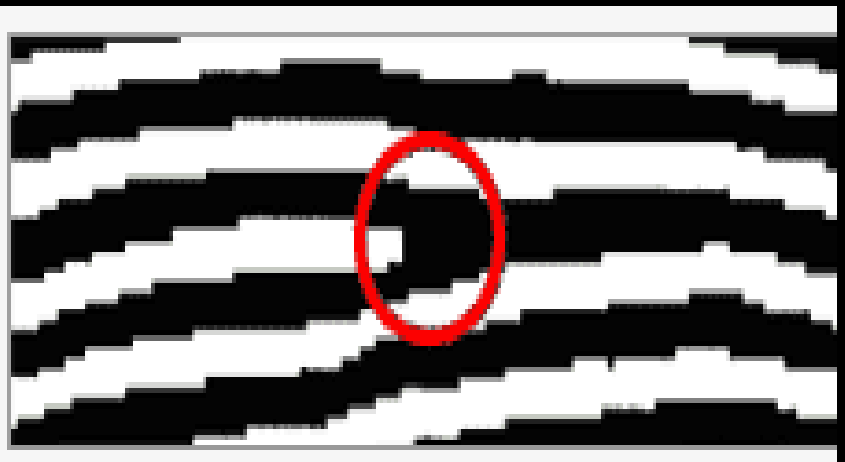
b. Ridge Bifurcation

Fingerprint Authentication

Discriminating Information



a. Ridge Ending



b. Ridge Bifurcation

Fingerprint Authentication

Authentication Procedure

Fingerprint Authentication

Authentication Procedure

Two essential procedures

- **Enrollment**
- **Authentication**

Fingerprint Authentication

Authentication Procedure



Fingerprint Authentication

Authentication Procedure



Image Acquisition

Classified roughly as

- **Optical**
- **Non-Optical**

Fingerprint Authentication

Authentication Procedure



Fingerprint Authentication

Authentication Procedure

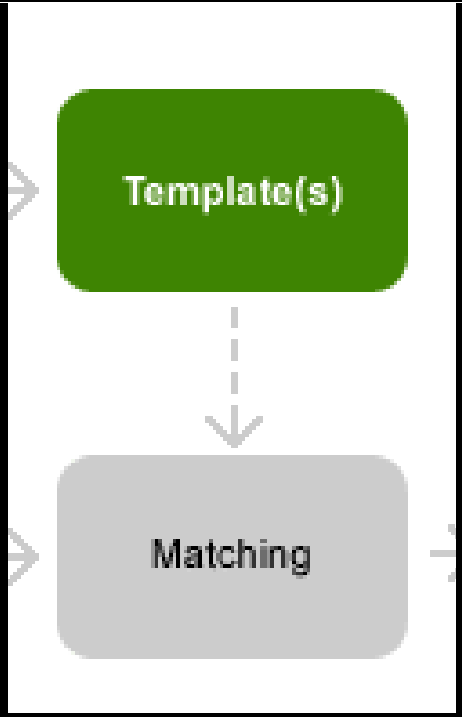


Feature Extraction

- **Direct**
- **Feature** or **Minutia** based

Fingerprint Authentication

Authentication Procedure

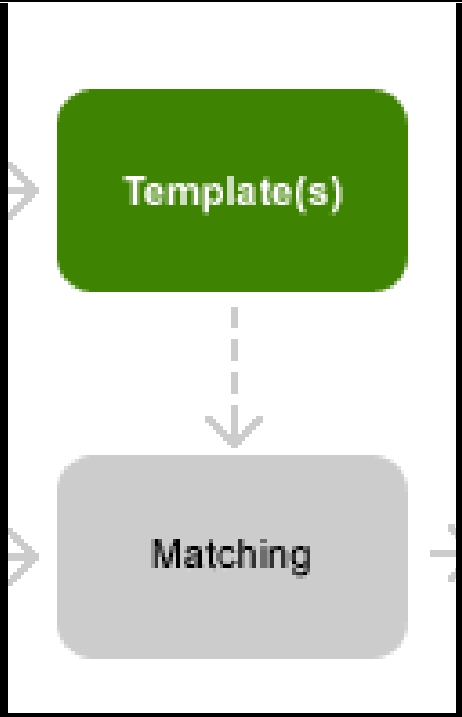


Fingerprint Authentication

Authentication Procedure

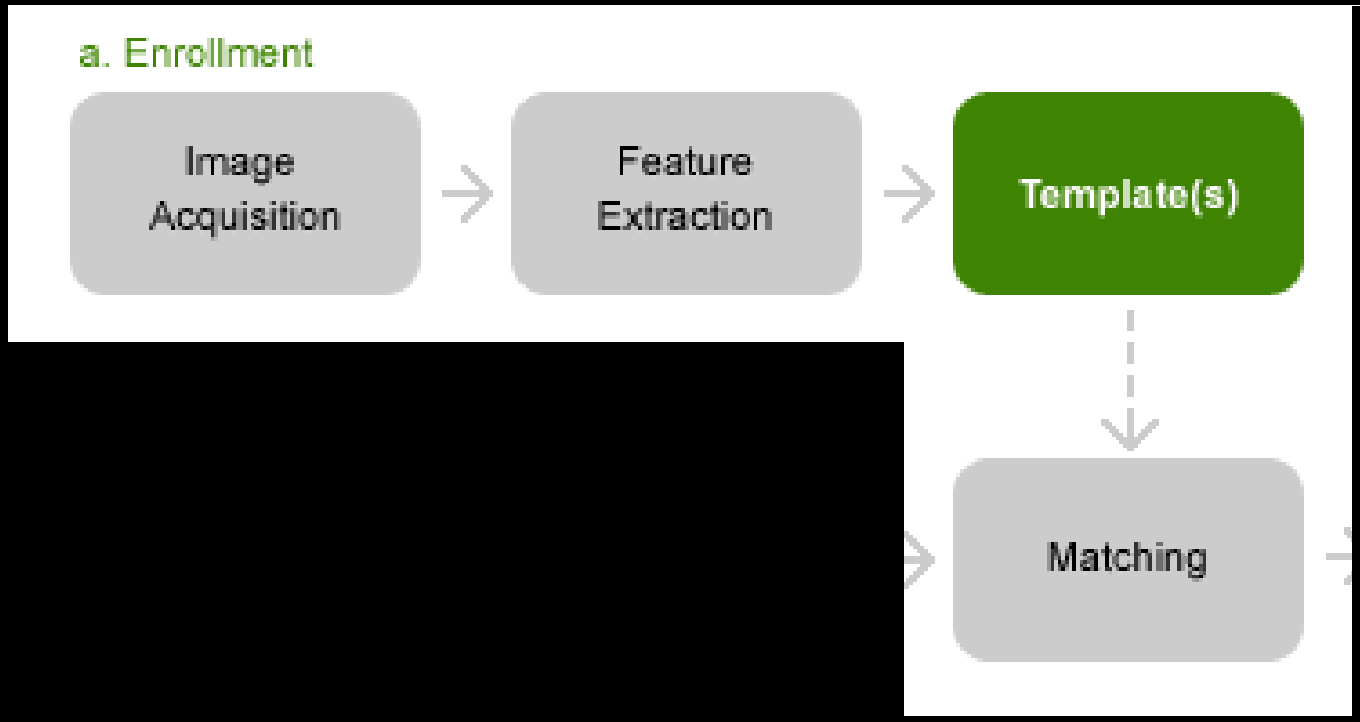
Matching

- 1 : 1 Matching
- 1 : N Matching



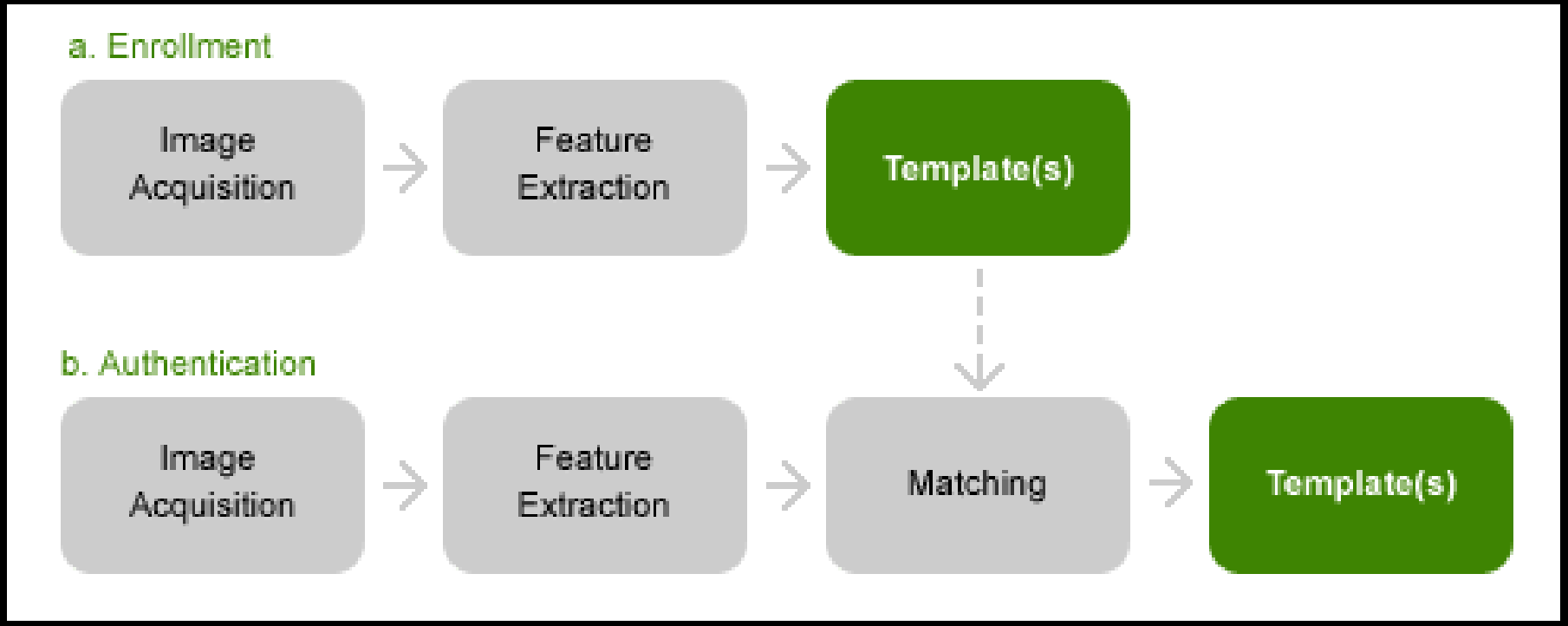
Fingerprint Authentication

Authentication Procedure



Fingerprint Authentication

Authentication Procedure



Fingerprint Authentication

Applications

Fingerprint Authentication

Applications



- Biometric Smart Gun

Fingerprint Authentication

Applications



- **Biometric Smart Gun**
- **Olympic Summer Games -** Athens, Greece 2004

Fingerprint Authentication

Drawback

Fingerprint Authentication

Drawback

- **Skin Diseases** prevent normal formation
- **Magali (Naegeli) syndrome**
- **The HuangTien family in Taiwan**

Fingerprint Authentication

Drawback



- **Skin Diseases** prevent normal formation
- **Magali (Naegeli) syndrome**
- **The HuangTien family in Taiwan**

Other Types

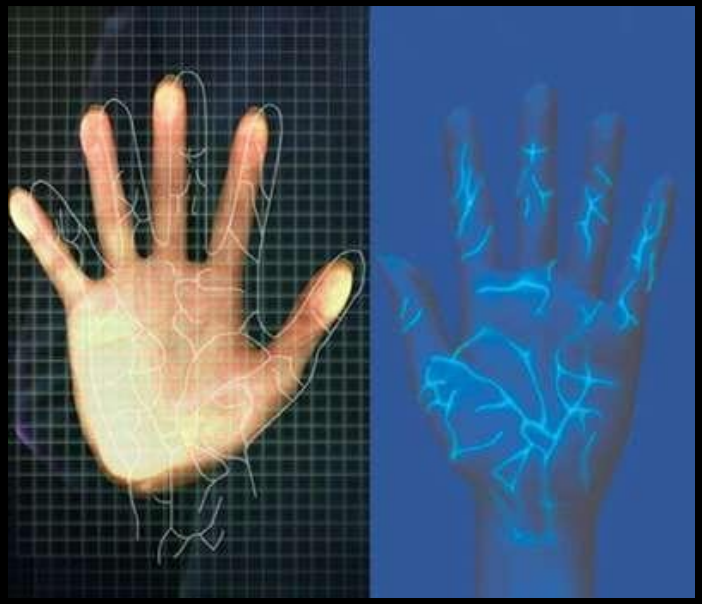
Other Types

Palm/Hand Geometry

Other Types

Palm/Hand Geometry

- **Full-Hand Geometry**



Other Types

Palm/Hand Geometry

- Full-Hand Geometry
- Two Finger Scan



Other Types

Palm/Hand Geometry

- Full-Hand Geometry
- Two Finger Scan



Image Acquisition

Other Types

Palm/Hand Geometry

- Full-Hand Geometry
- Two Finger Scan



Image Acquisition

Outline

Other Types

Palm/Hand Geometry

- Full-Hand Geometry
- Two Finger Scan



Image Acquisition | Outline | Geometry

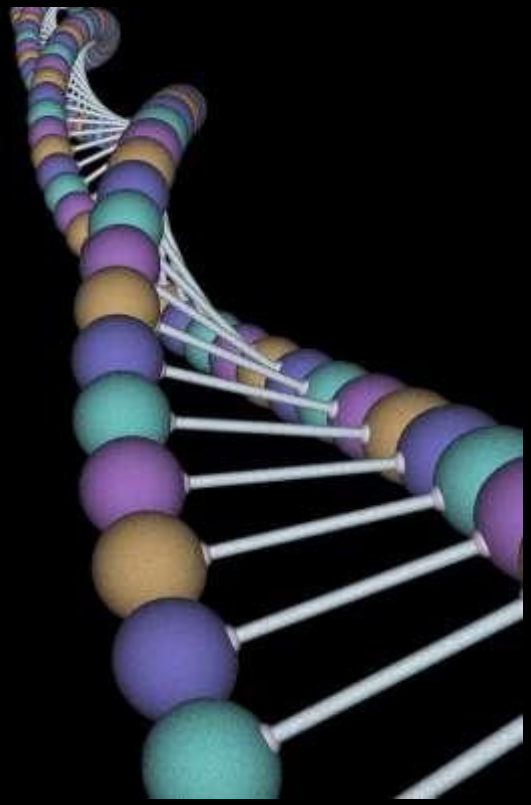
Other Types

DNA

Other Types

DNA

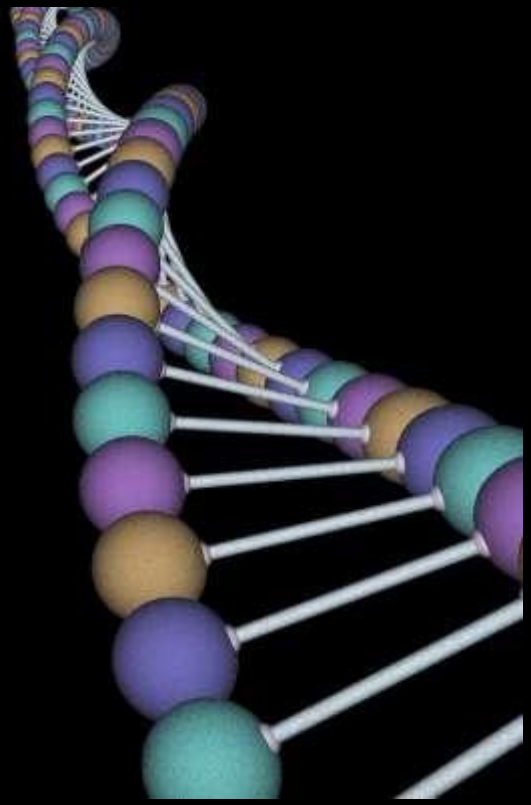
- Requires Actual Physical Sample



Other Types

DNA

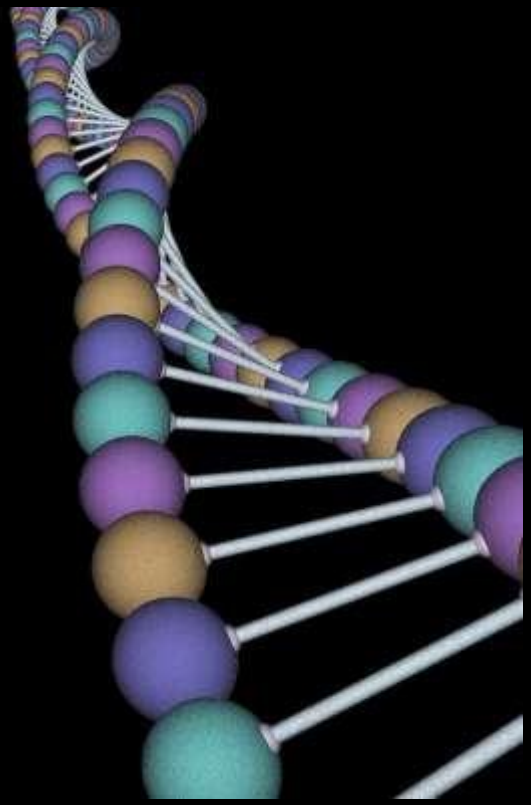
- Requires Actual Physical Sample
- Not fully automated



Other Types

DNA

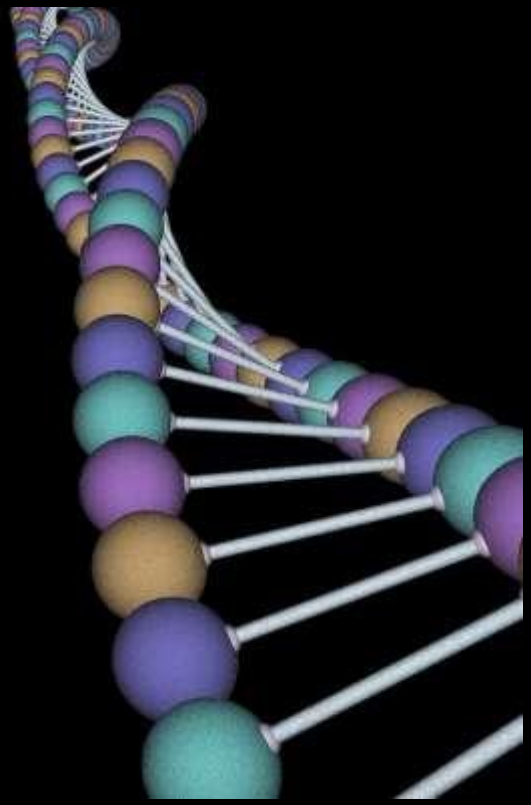
- Requires Actual Physical Sample
- Not fully automated



Other Types

DNA

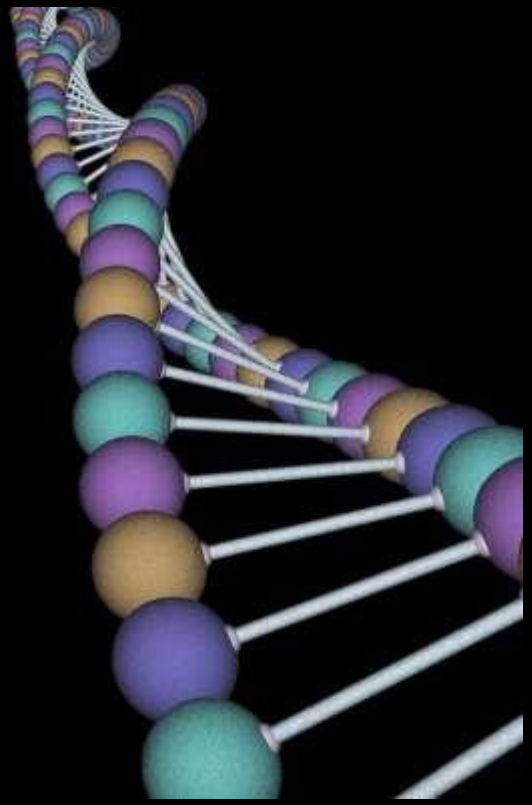
- Requires Actual Physical Sample
- Not fully automated
- Expensive



Other Types

DNA

- Requires Actual Physical Sample
- Not fully automated
- Expensive
- Privacy Issues



Other Types

Gait

Other Types

Gait

- Study of Animal Locomotion



Other Types

Gait

- Study of Animal Locomotion
- Helps Athletes



Other Types

Gait

- Study of Animal Locomotion
- Helps Athletes
- Standard Camera – any condition



Other Types

Gait

- Study of Animal Locomotion
- Helps Athletes
- Standard Camera – any condition
- Identification at a Distance



Other Types

Skin Reflection

Other Types

Skin Reflection

- Absorption Spectrum of Skin varies



Other Types

Skin Reflection

- Absorption Spectrum of Skin varies
- LEDs send Light into Skin



Other Types

Skin Reflection

- Absorption Spectrum of Skin varies
- LEDs send Light into Skin
- Photodiodes read reflected light



Future of Biometrics

The Past & The Present



Future of Biometrics

Security & Privacy

Biometrics in Action

Cost & Accuracy

Security & Privacy

Biometric Systems must be Fool-Proof

Security of Collected Biometric Data

Biometrics In Action

Immigration Control

Biometrics In Action

Immigration Control



Biometrics In Action

Transactional Authentication

Biometrics In Action

Transactional Authentication



Biometrics In Action

Computer Security

Biometrics In Action

Computer Security



Biometrics In Action

Smart Doors

Biometrics In Action

Smart Doors

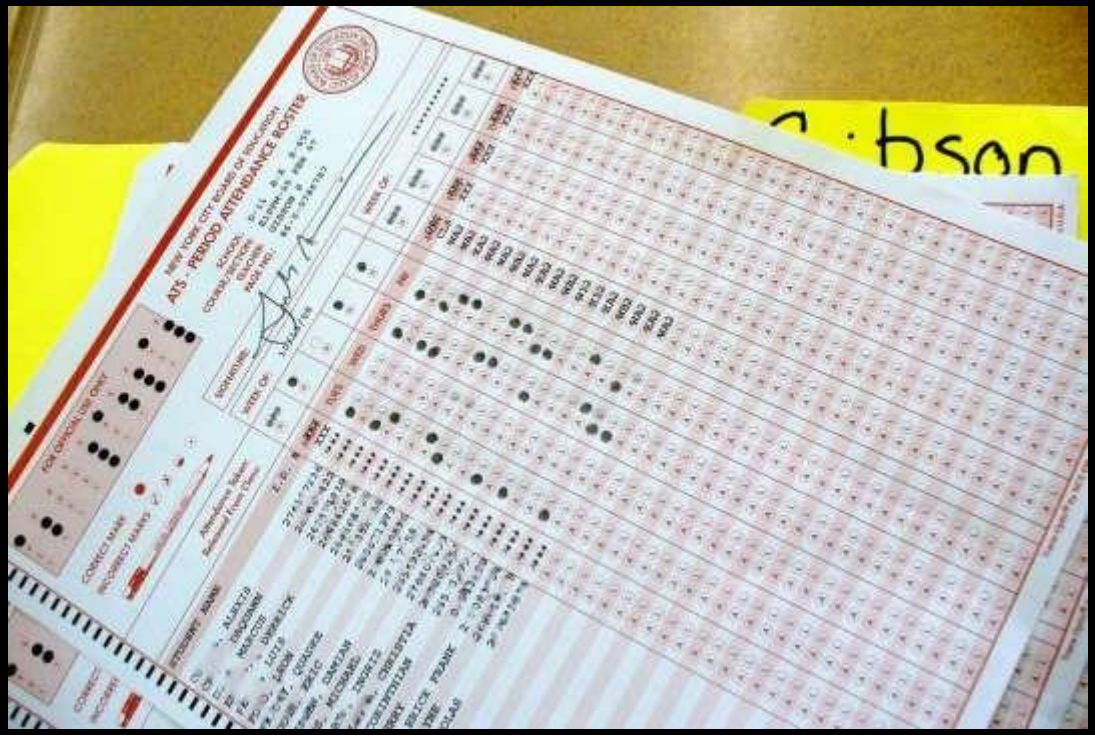


Biometrics In Action

Time and Attendance

Biometrics In Action

Time and Attendance

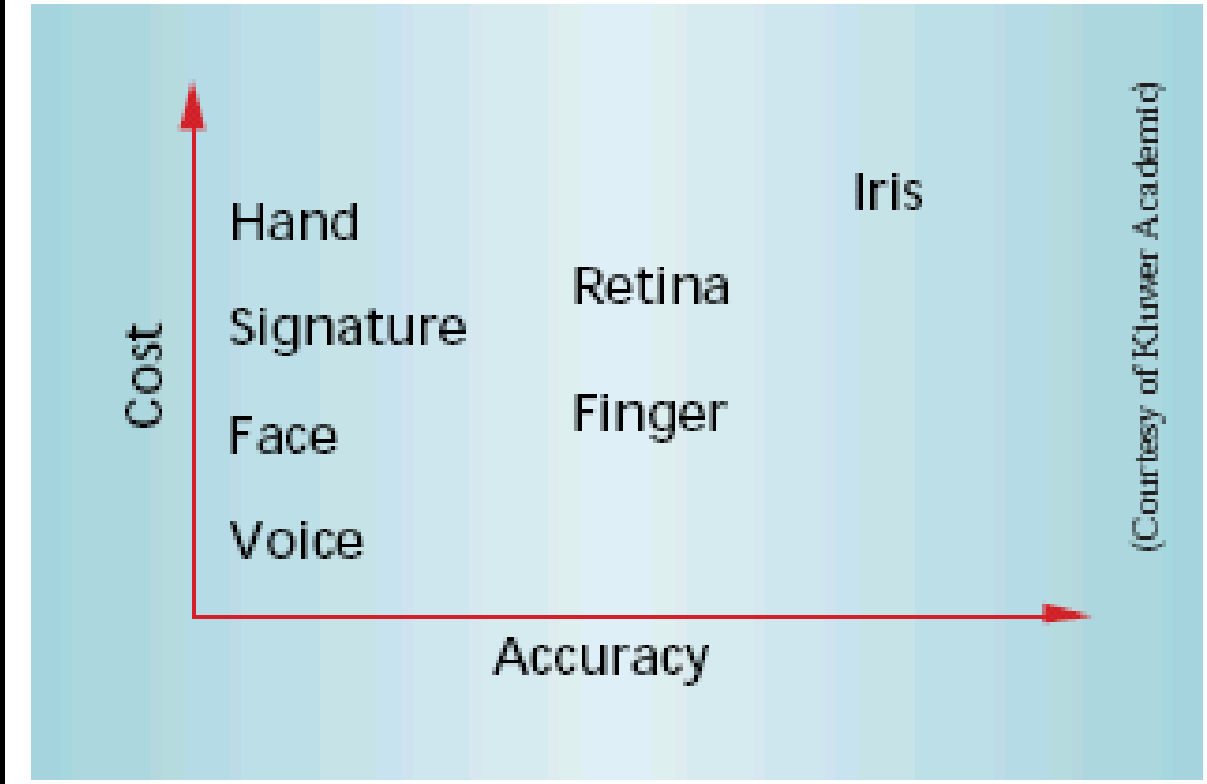


Cost & Accuracy

More Accuracy = More Cost

Cost & Accuracy

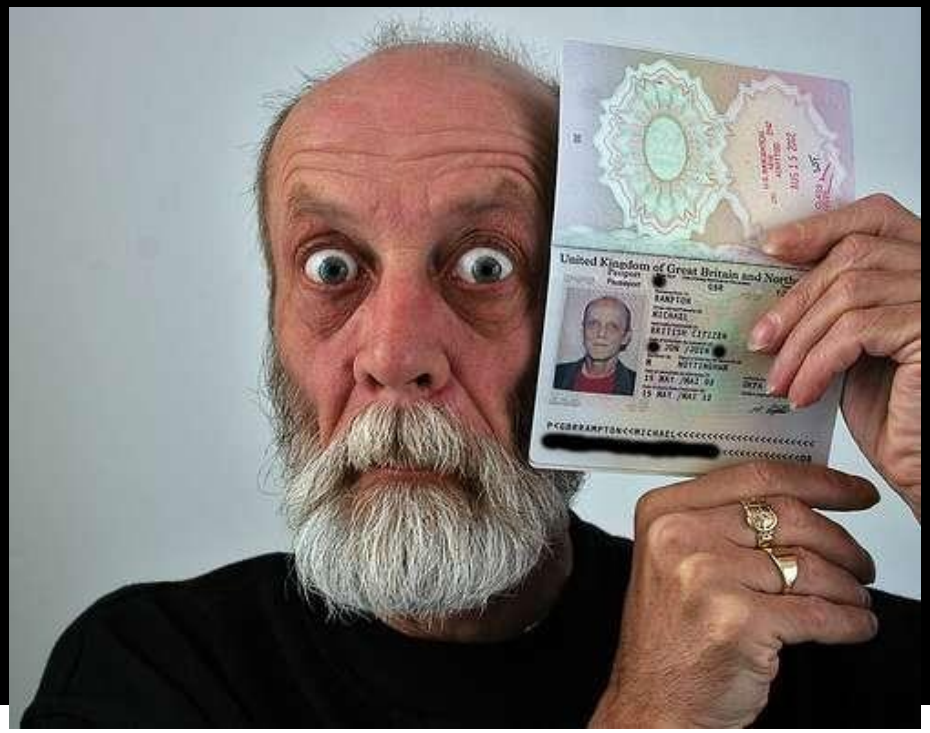
More Accuracy = More Cost



Conclusion

Conclusion

End of Paper based Identity



Conclusion

End of Identity Theft



Conclusion

Reduces Expenditure on Security

Saves Time

Helps us Fight Terrorism

Comfortable life + Improved Security

Conclusion

We need to -

- do Research in Biometrics
- bring Awareness amongst people
- Be ready for the change
- Bring about a change
- Become Smart

Further Reading

- Paul Reid . **Biometrics for Network Security** . Prentice Hall PTR. December 30, 2003

- Nalini KRatha . **Advances in Biometrics** . Springer . 2007

- **Biometrics Consortium** -
- http://www.biometrics.org/resources.php

- **Biometrics Products** –
http://www.biometricsproducts.com/

- **Introduction to Biometrics** -
http://biometrics.gov/ReferenceRoom/Introduction.aspx